



Deliverable C.2

“Impact assessment models”

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SUMMARY OF THE DOCUMENT	
The SIAMUS project (Integrated System for the Analysis of Sustainable Urban Mobility) aims to provide data and indicators for the sustainable urban mobility based on the experience of Pin Bike, the patented system for certification and gamification of sustainable mobility. This deliverable C.2 contains an in-depth study of Pin Bike initiatives and their impact on urban and sustainable mobility in cities where the initiatives have been active, concluded, or are currently active. The purpose of this document is to demonstrate, through a survey of all Pin Bike users in the urban areas considered in the "Integrated System for the Analysis of Sustainable Urban Mobility (SIAMUS)" project, the main impacts of the proposed and concluded Pin Bike initiatives. The objective is to develop proposed "impact evaluation models" based on mixed surveys, with both Stated Preference (SP) and Revealed Preference (RP) data, based on Pin Bike user experience during the initiative. These documents will provide a useful scientific basis for behavioral analyses and studies on travel mode choice, guiding future research towards the definition of active and sustainable mobility solutions through expected user behavioral changes. By analyzing end users' ex-post and ex-ante evaluations and considerations regarding potential initiatives to incentivize sustainable mobility, it will also be possible to conduct cost-benefit analyses on new project proposals or initiatives. In synergy with the technical report (Deliverable C.1), these models will be the basis for the development of guidelines for public and private policy makers (Deliverable D.1), the result of Activity D.	

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Introduction

Deliverable C.2 of the SIAMUS (Integrated System for the Analysis of Sustainable Urban Mobility) project aims to provide an in-depth analysis and scientific analysis of the impacts of Pin Bike initiatives, a patented, innovative digital solution that integrates certification and gamification to encourage sustainable mobility in urban areas. Specifically, this deliverable focuses on the effectiveness of these initiatives in changing users' mobility habits, promoting behavioral change toward more sustainable modes of transportation that are resilient to the environmental and urban planning challenges of cities.

The central objective of this paper is to analyze the system's impact, studying the effects, evaluating ex-post conditions, and analyzing ex-ante conditions in the urban areas affected by Pin Bike initiatives. The study focuses on nine Italian cities (Bari, Bergamo, Bologna, Lecce, Pescara, Sassuolo, Florence, Turin, Trento) where Pin Bike initiatives have been implemented, have concluded, or are currently active, with different incentive methods or characteristics.

Using an in-depth investigation based on empirical data collected through a questionnaire administered to system users, the impacts of these initiatives on urban mobility will be analyzed. This approach will aim to identify, quantify, and interpret the direct and indirect effects of Pin Bike on mobility habits and the perception of cycling as an alternative mode of transportation, as well as the economic, social, and environmental impacts of its use.

Specifically, deliverable C.2 is an analytical and evaluative document that analyzes the survey conducted within the SIAMUS (Integrated System for the Analysis of Sustainable Urban Mobility) project. The proposed survey is distinguished by its mixed methodology, with a strong emphasis on the use of stated preference (SP) and revealed preference (RP) surveys. These methodologies are used to gather information regarding not only users' explicit motivations for choosing transportation modes, but also the more subtle, and sometimes unconscious, dynamics that drive daily mobility behavior. By combining the two types of surveys (SP and RP), it is possible to identify potential specific user behaviors which, considering the nationwide reach of Pin Bike initiatives, could lead to interesting political and sociodemographic considerations.

Objectives of the study

The investigation carried out and the subsequent study conducted in this deliverable C.2 aim to:

- Analyse changes in urban mobility behaviour: Evaluating how the Pin Bike initiative has influenced users' travel choices, in particular the shift from the use of motorised vehicles towards active and sustainable transport modes, such as cycling.
- Evaluate the effectiveness of the Pin Bike incentive system: Examining the impact of gamification dynamics and certifications in motivating users to use greener modes of transport, including through the introduction of rewards and incentives linked to sustainable mobility behavior.
- Support future planning of urban mobility: By providing guidance to administrations and policy makers, useful for predicting new policies to incentivize the use of alternative transport modes, at the same time it provides details on how to optimize investments in sustainable urban mobility infrastructure.
- Analyse the type of travel: By differentiating the types of travel (systematic home/school and home/work trips, and generic trips for example for shopping) it is possible to determine important characteristics of urban mobility, both in terms of traffic management and from the point of view of the management of urban spaces.
- Provide a scientific basis for future behavioral research: By analyzing the results obtained, this deliverable will contribute to a greater understanding of the behavioral dynamics related to the choice of travel modes and will provide indications to guide future research towards active and sustainable mobility solutions.

Deliverable recipients

This document is primarily aimed at two audiences: the scientific community studying and analyzing mobility phenomena and future solutions, and public administrators and policymakers, both at the local and national levels, involved in city planning and urban mobility management. The aim is to provide a useful methodological and scientific framework to guide future policies, with the goal of integrating active and sustainable mobility solutions in increasingly congested and polluted urban contexts.

Outside of the project, the following document may also have an impact on mobility sector stakeholders (e.g., public transport companies, shared mobility service providers, and private companies), who could benefit from the application of the proposed impact assessment models and guidelines. These stakeholders can use it as a basis for implementing innovative solutions that encourage the use of more sustainable means of transport.

The main purpose of the deliverable is therefore to provide empirical evidence on the effectiveness of the Pin Bike system, to support the evolution of sustainable urban mobility through policies that encourage a change in user behavior, promoting the transition towards greener and more inclusive transport models.

Scientific contribution

The scientific contribution of this deliverable lies in the in-depth analysis of a qualitative behavioral survey administered to a large number of users, with different socioeconomic characteristics, from different urban contexts, all interested in transport modes active and sustainable and active users of Pin Bike.

The scientific validity of the analyses conducted on the surveys developed and administered to Pin Bike users is demonstrated by the sample size, which is representative of the entire user population. Considering the nine cities analyzed, we counted approximately 7,000 users in total who were administered the questionnaire.

Considering a confidence level of 95% with a margin of error of 5% the sample size should be 365. In total we received 517 responses from the submitted questionnaire.

Total responses, primarily drawn from the cities of Bari, Bologna, and Florence. It should be noted that in seven of the nine cities included in the SIAMUS project, the Pin Bike projects have concluded. Initiatives are still active in Florence and Bologna, and in Bologna in particular, two projects are active each for the AUSL and AOSP. We can therefore state that the sample size, the responses recorded, and the analyses performed are representative of almost the entire population studied.

Added to these analyses are those carried out on the historical Pin Bike questionnaires for various projects, especially for the cities of Bergamo, Trento, Turin and Lecce, from which very interesting results are obtained. interesting.

The analysis conducted is useful for understanding the motivations and determining characteristics of users' sustainable mobility behavior. In particular, the questionnaire represents a crucial tool for collecting direct and specific data on the effects of Pin Bike initiatives on users' transportation choices.

The questionnaire is structured to collect a wide range of information, including:

- The reasons behind the choice of transport (e.g. sustainability, speed, comfort).
- The temporal variables and the average distances travelled by users.
- The barriers (infrastructural and otherwise) perceived in using the bicycle as an alternative to the car.
- The frequency of use of the service and the evaluation of the overall experience.

This data will provide a detailed overview not only of actual changes in users' mobility behavior, but also of individual perceptions regarding the sustainability, safety, and convenience of different means of transportation.

The importance of the questionnaire lies in the fact that it provides an objective picture of users' experiences and opinions, allowing us to validate or invalidate hypotheses about behavioral dynamics related to the adoption of sustainable mobility solutions and provide practical guidance for optimizing incentive policies. The approach of asking questions based on stated preference (SP) and revealed preference (RP) generates a comprehensive survey that is easily applicable to all user categories, minimizing errors in identifying the objectives by the respondent.

In summary, Deliverable C.2 represents a significant opportunity to advance the scientific understanding of the factors that influence urban transportation mode choice in different contexts, with initiatives to encourage active mobility. This is highly useful for guiding future mobility strategies, integrating digital solutions like Pin Bike into a long-term sustainability framework. The user questionnaire thus becomes a key element in building predictive mathematical models and in policy decisions aimed at improving the quality of mobility in cities.

Proposed questionnaire

The questionnaire administered to Pin Bike users is a crucial element of Deliverable C.2 of the SIAMUS project. The questions were designed to collect relevant data regarding users' mobility experiences before and after using the Pin Bike system. Below, an overview of the questions posed in the questionnaire is provided, along with an in-depth analysis of their purpose and the useful insights that can be gleaned from each.

The survey developed and analyzed was conducted in nine Italian cities: Bari, Bergamo, Bologna, Lecce, Pescara, Sassuolo, Florence, Turin, and Trento. The questionnaires were analyzed to determine the questions posed to all project users, and the responses to the questionnaires were subsequently analyzed to validate the answers. It was found that for all projects whose impact was analyzed, the questions were consistent with the objectives of the survey conducted and proposed to analyze the impact of the Pin Bike system on sustainable urban mobility.

The questionnaire was submitted through the dedicated section in the Pin Bike app. This sent users a notification with the questionnaire to complete. This enabled the user to be identified as participating in the survey using their User ID, which was obtained during registration for the Pin Bike app. The User ID was used by default to retrieve general information about the user registered in the app, such as date of birth, gender, city and place of residence, and city and place of work. Therefore, no questions related to these characteristics were included in the questionnaire.

The questionnaire structure will be shown below, along with some of the questions asked to determine the rationale behind the survey. The questionnaire consists of a series of sections, discussed below, with clearly defined objectives. The order of the questions is not random, but is useful for making their understanding more fluid and guiding the users towards a more accurate response.

General travel habits

Questions in the survey:

- *“What is your average daily commute to school or work?”*
- *“What time slot do your general travels occur in?”*

This section collects general travel information, such as distance, time, and time slot. Generally, we can consider two approaches: the user does not change these habits in relation to the initiative or a possible modal shift, or the user changes these habits by virtue of a modal shift or due to other factors. These considerations are very

Important. In this phase, the distance between home and work/school is recorded to analyze the spatial context of mobility and the adaptation of the Pin Bike system to user needs. The distance in kilometers between home and work/school is also recorded. This is very useful in analyzing systematic trips. This allows us to determine whether the Pin Bike initiative is more effective in densely populated urban areas or in contexts where the distance traveled is significant. The question about the time slot helps identify the periods of greatest congestion or transportation need during the day, allowing us to understand whether Pin Bike has an impact on the management of rush hour and the temporal distribution of trips.

Behavioral habits and characteristics of travel before Pin Bike Questions in the survey:

- *“Before the Pin Bike project, how did you get around the city?”*
- *“Before Pin Bike, what were the reasons behind choosing this mode of transportation?”*
- *“Before Pin Bike, how long did it take you on average to travel from home to school?
or home-work?”*

This section of the questionnaire, using these questions, aims to collect data and information on the transportation modes and trip characteristics used by users before joining the Pin Bike initiative. It also explores the motivations that drive users to choose certain transportation modes, providing a fundamental context for analyzing subsequent behavioral change. These questions are essential for understanding users' pre-existing urban mobility behavior.

In the first question, the data allows us to identify the most used mode of transport (car, public transport, bicycle, etc.), to analyse whether the Pin Bike initiative has had an impact.

significant in reducing the use of motorized vehicles and increasing the adoption of more sustainable modes such as cycling. The answers to the second question help us understand which factors/attributes (speed, safety, convenience, sustainability) are prevalent in transportation modal choice. The question of travel time is important for analyzing and comparing pre- and post-modal shifts to determine whether travel time decreases or increases with a modal shift. This analysis provides a useful starting point for comparing any improvements in travel speed after the adoption of Pin Bike and allows us to understand the extent to which travel time influenced transportation mode choice and whether the Pin Bike initiative had an impact on reducing travel times.

Changes recorded thanks to the Pin Bike initiative

Questions in the survey:

- *"With or after Pin Bike, what are the reasons behind the choice of travel mode?"*
- *"Since you started using Pin Bike, how much time do you spend or have you spent on average to travel between home and school or home and work?"*

This section investigates any changes in the motivations that influence transportation choices, comparing the pre- and post-intervention situation. The question features and options are the same as in the previous section, allowing users to assess any perceived changes in the same way. The data collected here allows us to observe how the Pin Bike initiative has influenced users' priorities (for example, an increase in awareness of environmental sustainability). Comparing travel time before and after the introduction of Pin Bike provides a direct measure of the initiative's effectiveness in improving transportation efficiency. The data allows us to observe whether using Pin Bike has reduced travel time and whether cycling has proven to be an efficient alternative to other modes of transportation.

Using Pin Bike

Questions in the survey:

- *"What inspired you to participate in Pin Bike?"*
- *"How often have you used the Pin Bike service?"*

In the transition to sustainable mobility, incentive measures like Pin Bike are "soft" measures. Unlike "hard" measures like the creation of dedicated infrastructure like bike paths or bike stations, they only show their effects when there is significant user involvement, especially in terms of a "willingness to change." This section analyzes the motivations that lead users to sign up for the Pin Bike initiative and use it. The following are analyzed:

The individual motivations that drive users to participate in the Pin Bike initiative, identifying the factors that most stimulate the adoption of sustainable mobility. The information gathered allows us to understand which factors (economic incentives, health, sustainability) were decisive in program participation, providing useful insights for improving future incentive initiatives.

By analyzing how frequently users used the Pin Bike service, allowing us to distinguish between regular and occasional users, it is possible to derive indicators of the program's effectiveness: a low frequency could suggest practical or motivational barriers to cycling.

Infrastructure assessments in bicycle use

Questions in the survey:

- *“On a scale of 1 to 5, how easy, comfortable, and safe was it to cycle to your destination?”*
- *“Were there parking facilities for your bike upon arrival?”*

Here, users can examine and evaluate certain characteristics of the infrastructure they are using. User perceptions of the convenience, safety, and efficiency of cycling—key factors for promoting the transition to sustainable mobility—are assessed using a Likert scale of 1 to 5. This allows for the identification of potential difficulties and obstacles to cycling, such as the lack of adequate infrastructure or safety issues, which is very useful information for policymakers during the planning phase. The analysis also focuses on bicycle parking infrastructure, such as bike racks and parking spaces. If the lack of support services (e.g., bicycle parking) is a deterrent to continued cycling, dedicated infrastructure can be developed. This information is also very useful for businesses, commercial activities, or local politicians, particularly with a view to promoting local economic growth.

Economic incentives

Questions in the survey:

- *“What new economic incentives would you introduce?”*
- *“Without financial incentives, would you still continue cycling?”*

Economic incentives are a factor that push many users to change their transportation mode in response to incentive projects. It's helpful to gather suggestions from users on how to improve economic incentives for sustainable mobility. This section offers insights into which incentives are considered most attractive and practical, guiding future incentive policies. Conversely, there's a risk that incentivized users will stop using sustainable transportation modes at the end of the project, returning to their previous habits. For this reason, it's important to focus on change.

Behavioral rather than purely financial compensation. At this stage, the user is also asked how likely they are to continue cycling without financial incentives. This stated preference question is important as it measures the effectiveness of financial incentives and users' propensity to maintain cycling as a means of transportation. This section is therefore very useful in ensuring that the financial resources invested in incentive projects are not wasted.

Shopping by bike

Questions in the survey: •

“Do you shop when you travel by bike?”

- *“What services and products do you prefer to purchase when you travel by bike?”*

Pin Bike is based on purchases made by bicycle, promoting local trade and encouraging a "new" way of using the bicycle as an alternative to the car, also with a view to

Purchases. This section was designed to examine the interaction between bicycle use and consumer behavior, which is important for understanding whether cycling can become a means of supporting local economic activities based on direct user experiences. The actual experiences revealed by users help us understand whether bicycle use influences purchasing behavior.

also in terms of purchased materials and whether there are opportunities for synergy between sustainable mobility, local policies and actions by commercial activities.

Post-initiative perception of cycling, project evaluations, and behavioral change

Questions in the survey:

- *“Thanks to Pin Bike, how much has the perception of the bicycle as a mode of transport changed?
alternative to the car?”*
- *“Evaluate the overall project experience”*
- *“Would you participate in the Pin Bike project again?”*
- *“Suggestions for improvement”*

The final phase of the investigation involves a post-initiative behavioral evaluation, analyzing how much the user actually perceives a change in their travel habits.

Measure the effectiveness of the Pin Bike program in changing users' perceptions and attitudes toward cycling. The data on bicycle perception provides a crucial indication of the initiative's success in influencing transportation behaviors and preferences. Likewise, the level of user satisfaction with the entire program, subsequently measuring their willingness to

to join future similar programs. Finally, useful feedback is collected to improve the program, providing ideas and solutions to address any issues encountered.

The proposed questionnaire is designed to collect comprehensive and exhaustive data on the effects of Pin Bike initiatives, not only on travel methods but also on user perceptions, motivations, and behaviors. Each question provides valuable information for analyzing the project's effectiveness and guiding future sustainable mobility strategies.

In the next phase, the results of the survey will be shown.

Case study analysis and results

This section shows the results of the survey administered to Pin Bike users of the included projects. Socioeconomic variables such as age and gender were not explicitly requested in the survey but were instead retrieved by the system via the Pin Bike user ID with which they are registered in the app.

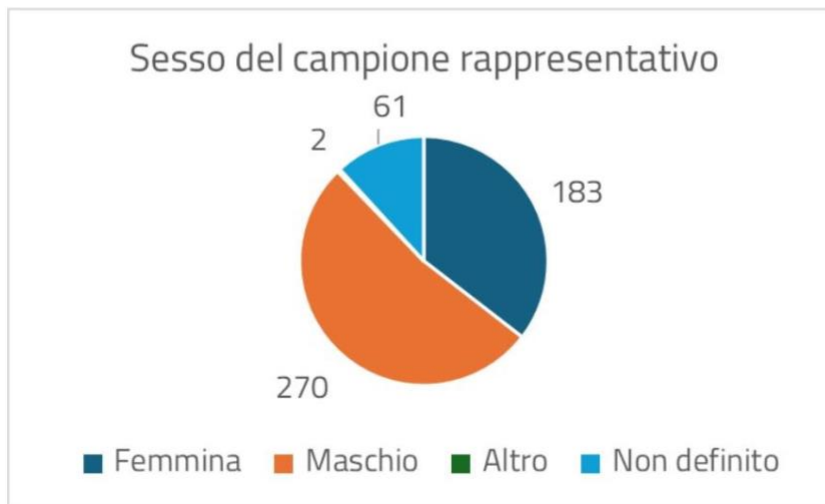
From a sample characterization perspective, we found an average age of 47. Below is a graph showing ages, grouped into 5-year ranges, which shows that the majority of users fall between 51 and 56 years of age, indicating that adult users perceive the bicycle as an alternative means of transportation to the car. Among younger users, this perception declines, dropping dramatically under the age of 30.

Chart 1: Average age of the representative sample



Regarding gender, approximately 52% are male and 35% are female. This data reflects how men value the bicycle as a means of transportation more than women. It has been further analyzed that this perception is due to women's perceived lack of safety.

Chart 2: Sex of the representative sample (female, male, other, not defined)



The users who participated in the survey were cyclists who began using bicycles thanks to Pin Bike and the initiatives launched in various urban areas. There are also users who were already commuting by bicycle before the incentive initiatives began. Approximately 43% stated they were cyclists even before the Pin Bike project, and increased their cycling after the initiative.

Chart 3: Before the Pin Bike project, how were you moving in your town? (public transport, walking, motor scooter, private scooter, intermodality, motorbike, bike sharing, private bike, car)

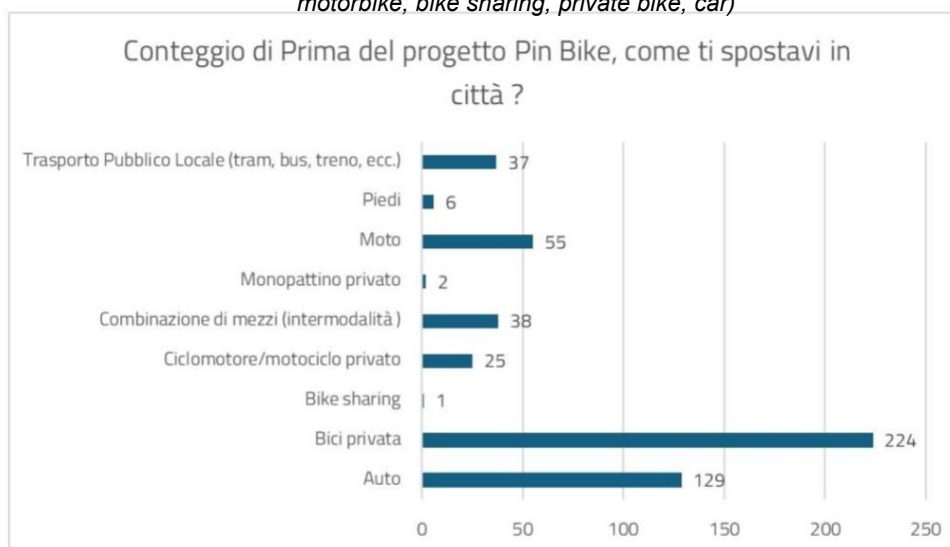
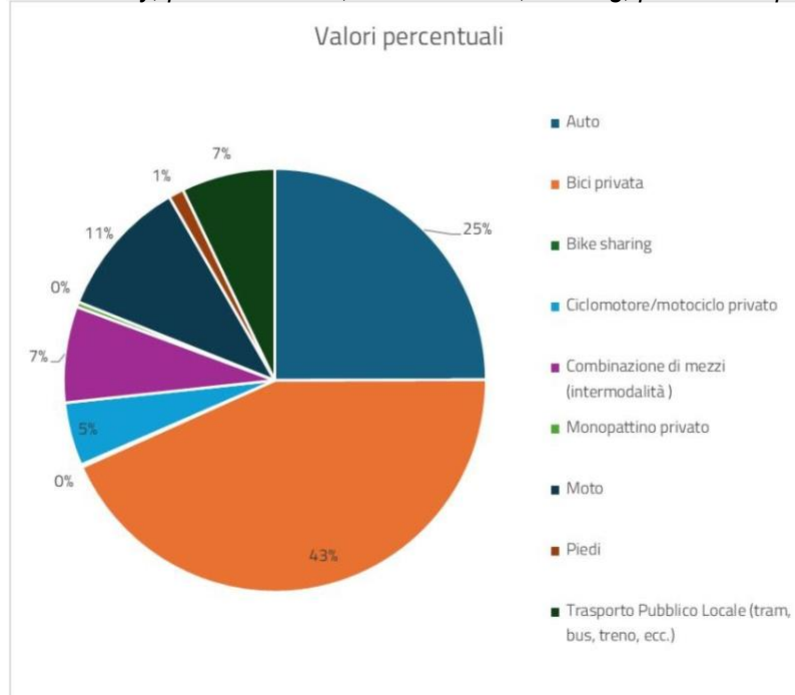


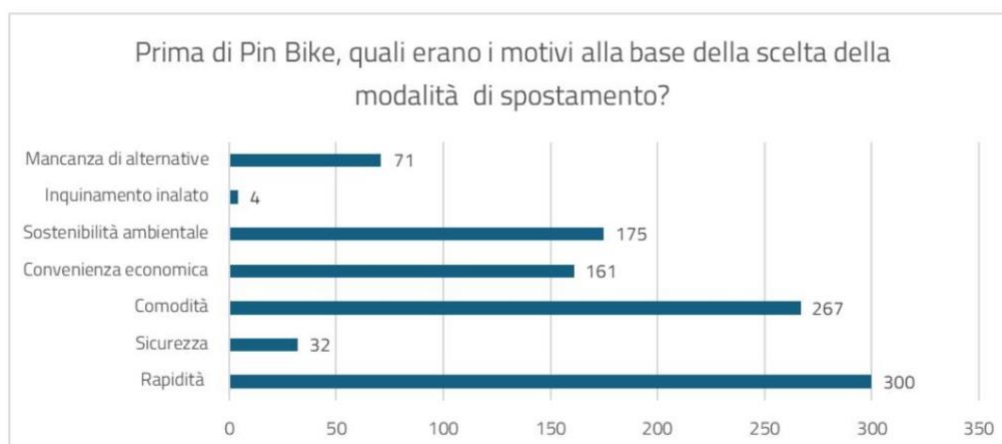
Chart 4: Percent values (car, private bike, bike sharing, motorcycle, intermodality, private scooter, motor scooter, walking, public transport)



Note that 25% of users used the car for their travel in urban areas of cities, despite the fact that all cities had alternative services such as Local Public Transport, a mode used by only 7% of total users.

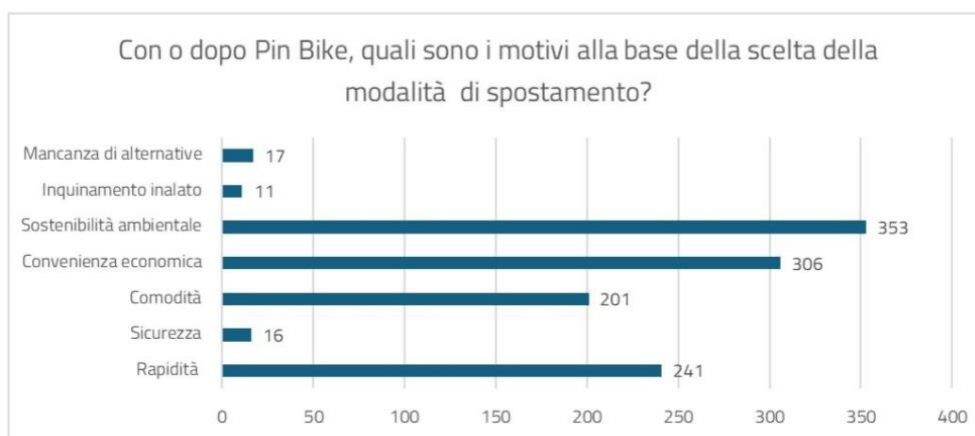
The reasons for choosing travel are varied: 30% of users based their choice on speed, followed by convenience at 26%. It's sobering to note that inhaled pollution accounts for less than 1% of the total travel time, and was only chosen as a factor in modal choice four times. This aspect is extremely important from a public health perspective, especially for users who travel on foot, by bicycle, by micromobility systems, or by motorcycle, where there is greater exposure to urban pollutants.

Chart 5: Before Pin Bike, what were the reasons of your modal choice? (lack of alternatives, inhaled pollution, environmental sustainability, economic benefit, comfort, safety, speed)



These assessments change with the Pin Bike initiatives. We analyzed how during and after the Pin Bike projects, users' motivations for choosing different modes of transport changed, increasing the choice of more environmentally sustainable modes of transport from 17% to 31%, and more economically convenient modes of transport from 16% to 27%. These considerations reduce the speed of travel, which drops to 21%, and convenience to 18%. The explanation for these phenomena is certainly the presence of economic incentives, which push users to use modes of transport that offer the possibility of receiving the rewards indicated in the various Pin Bike initiatives, but also the different perception of travel, less "frenzy" in the search for the fastest journey and greater attention to parameters environmental and economic.

Chart 6: With or after Pin Bike, what are the reasons of your modal choice? (lack of alternatives, inhaled pollution, environmental sustainability, economic benefit, comfort, safety, speed)



Travel time clearly remains one of the primary factors in choosing a trip, both modally and in other ways, such as choosing the route for privately operated systems. Overall, most users spent less than 30 minutes on their regular home-school and home-work trips. Approximately 46% fell within this time range, while the remaining 52% was split primarily between those who spent less than 15 minutes and those who spent less than an hour. Just over 2% of the total sample in urban areas spent more than an hour on their home-school or home-work trips.

This situation is almost identical for users during the Pin Bike projects. This indicates that the temporal impact of travel is due to a series of factors and that, although some users have changed their travel methods, travel times have not changed.

Chart 7: Before Pin Bike, how long were you commuting for? (More than 1h, less than 1h, less than 30 min, less than 15 min)

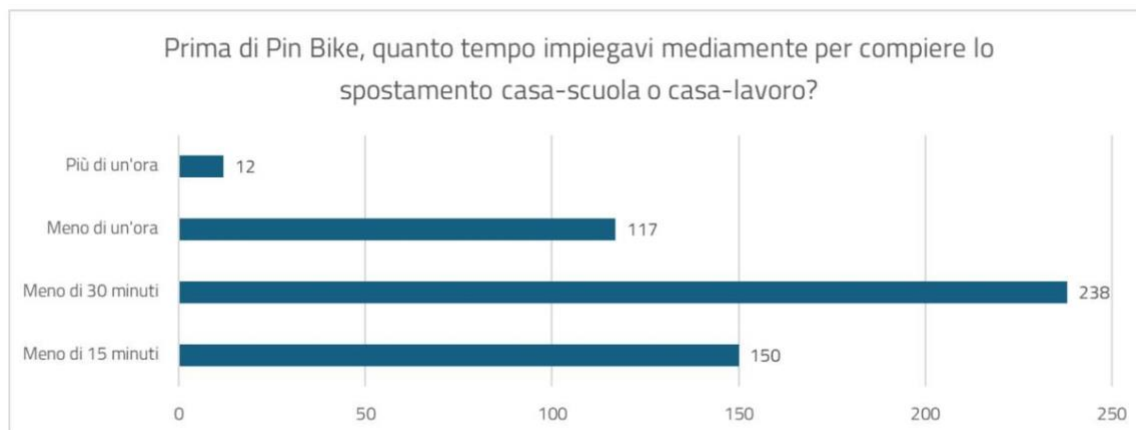
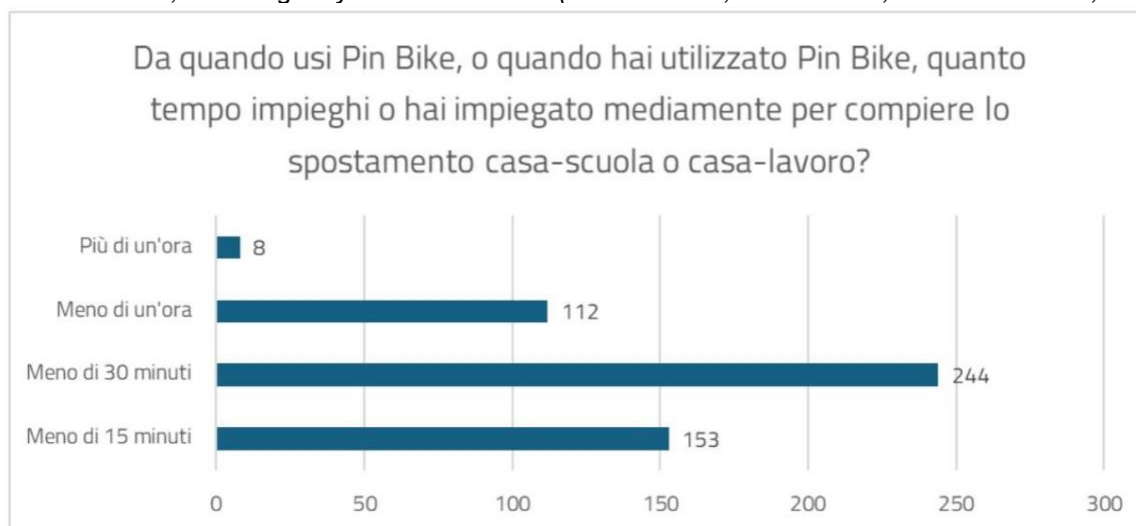
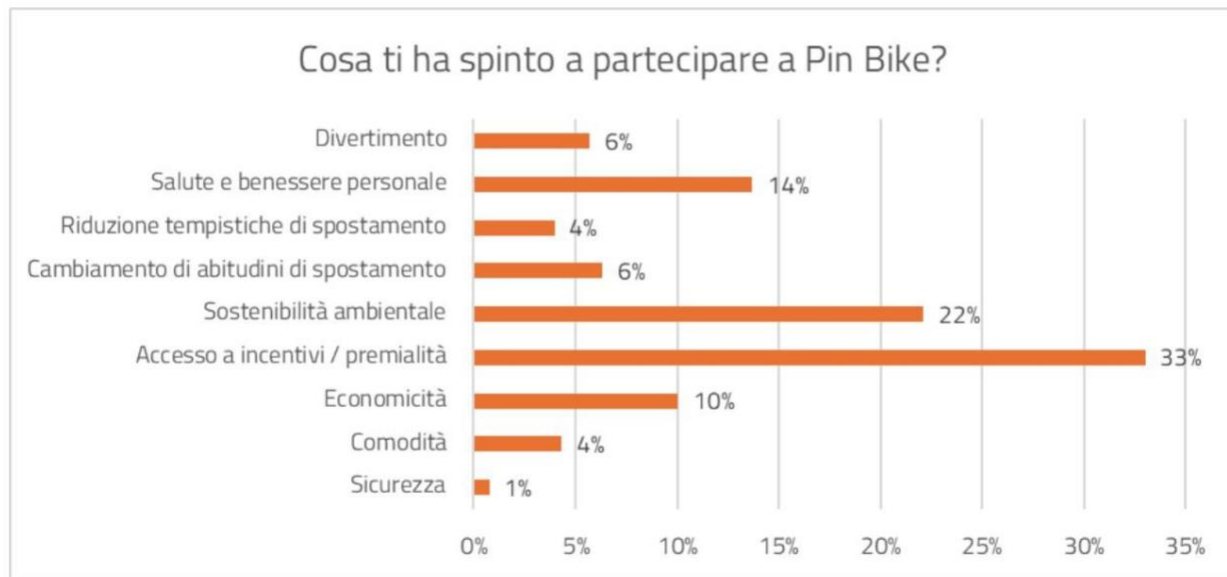


Chart 8: Since Pin Bike, how long do you commute for? (More than 1h, less than 1h, less than 30 min, less than 15 min)



Quantitatively, this translates to an average of approximately 7.51 km traveled within an urban area for a single systematic trip between home and school or home and work. Specifically, some users in the city of Bologna, who work from the local health authority (AUSL), travel more than 30 km within an urban area, reaching up to 55 km. In general, however, we can note that Florence is the city with the highest average travel distance for systematic trips, with more than 10% of users making a single trip of more than 15 km.

Chart 9: What made you use Pin Bike? (Fun, welfare, reduced time for commuting, habit change, environmental sustainability, incentives, affordability, comfort, safety)

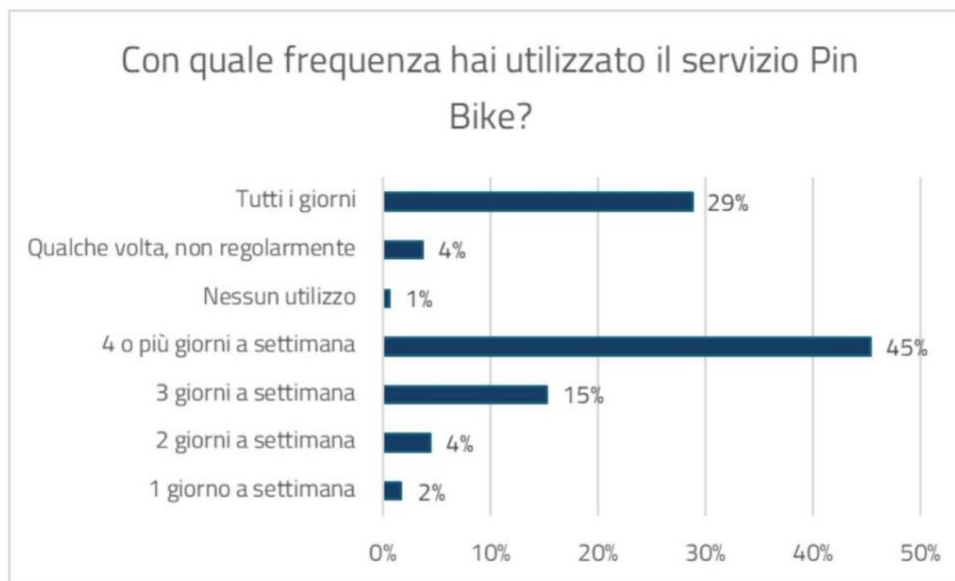


It's therefore important to analyze how to encourage more users to participate in initiatives promoting sustainable mobility. Having direct feedback on what motivated users to sign up for Pin Bike and participate in city projects is certainly a way to analyze past and current initiatives and plan new ones as effectively as possible.

33% of the sample indicated that access to financial and other incentives and rewards is an important driver for participation in such projects.

These results may also be useful for other evaluations. Reducing travel times is a parameter not strictly linked to Pin Bike initiatives or initiatives encouraging the use of more sustainable modes of transportation, although, as indicated in the final section of the survey, where users could submit suggestions and improvements, it could be a key factor in the use of alternative modes of transportation to the car. This is where intervention is certainly possible from a planning and policy perspective. Many users suggested increasing and improving the management of cycle paths and infrastructure, declaring them often dirty or unusable due to the presence of shrubs and litter. Other users also suggested that encouraging travel along cycle paths would lead to greater use of the infrastructure, improving its perception by the entire community, and "freeing up" roads shared by cars and buses.

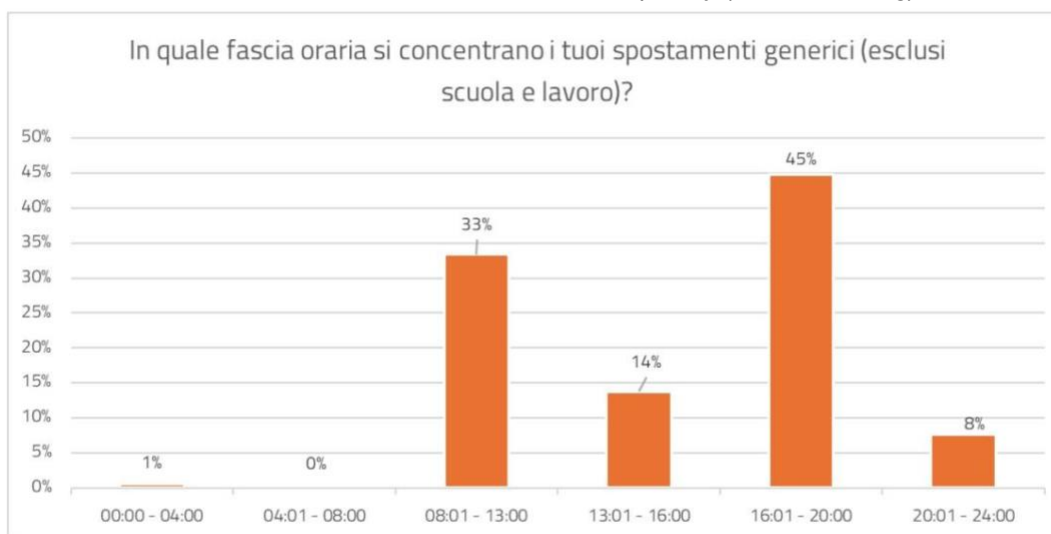
Chart 10: How frequently have you used Pin Bike? (Everyday, sometimes, never, 4+ d/w, 3d/w, 2d/w, 1d/w)



In total, approximately 75% cycle every day or at least 4 days out of 7. This means that all Pin Bike users, even those who generally used the car for their journeys before the start of the projects, have started cycling. Only a small number of users use Pin Bike a few times a week or never, and users are often inactive or have never recorded sessions in the various initiatives.

General travel plays a fundamental role in the volume of weekly trips. In particular, it has been analyzed how, within a single day, a single user travels and moves in different time slots of 4 hours each, dividing the 24-hour day into 6 distinct slots.

Chart 11: In which timeslot do travel more frequently (not commuting)?



Generic movements and systematic movements are concentrated essentially in 2 daily bands, from 8.01am to 1.00pm as the main morning band, and from 4.01pm to 8.00pm, the morning band.

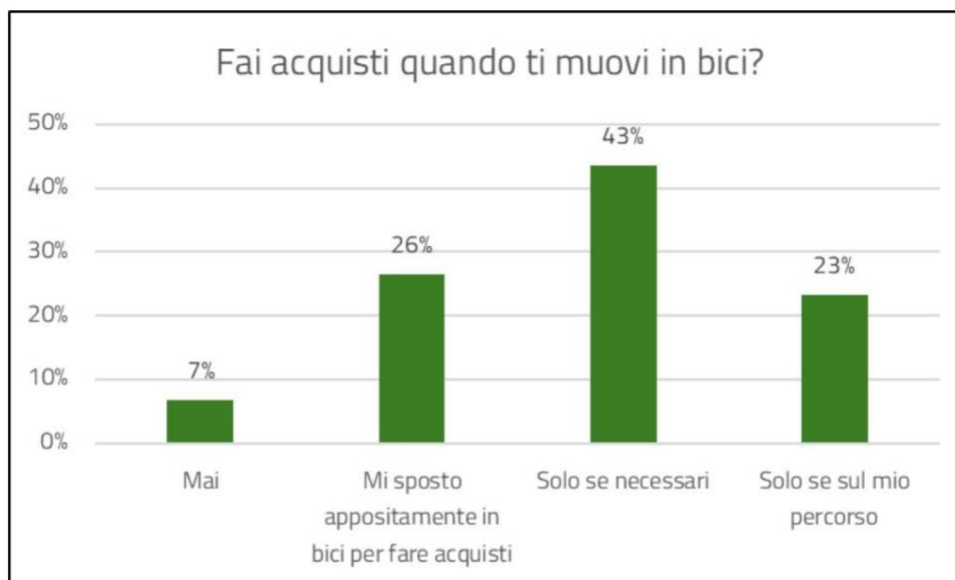
Main afternoon. This decreases dramatically in the early afternoon and evening hours after 8:00 PM. These considerations can be further explored by narrowing the time window of each time slot and specifically analyzing different conditions. For consistency, these timeframes were considered in relation to the different conditions and conformations of the urban areas of the projects analyzed.

Overall, the analyses conducted reflect a sample of users satisfied with the ease, convenience, and safety of cycling from origin to destination. The Likert scale analysis shows an average score of 4.1 out of 5, indicating that in recent years, urban areas have also been placing greater emphasis on the availability of bicycle services, such as bike parking, racks, and related infrastructure, as evidenced by approximately 300 users participating in the survey.

Among the various types of travel, much attention was paid to shopping trips. In particular, since the Pin Bike projects also involve trips to urban merchants, thus encouraging local commerce, attention was paid to how users perform this type of trip in order to understand its qualitative impact.

Users were primarily asked whether they make any trips when they cycle.

Chart 12: Do you go shopping while cycling? (never, purposely, only if necessary, only if on my way)



Overall, it was observed that 43% of users make purchases only out of necessity, while 26% of users use their bicycles specifically for transportation and shopping. This type of travel is directly related to other urban factors and problems, such as parking areas, which are often

Relatively small, paid areas, and central city areas with shops and businesses within the ZTL (limited traffic zone). In general, after analyzing the main types of products purchased while cycling, it emerges that users generally cycle to purchase food products several times a week, as previously noted.

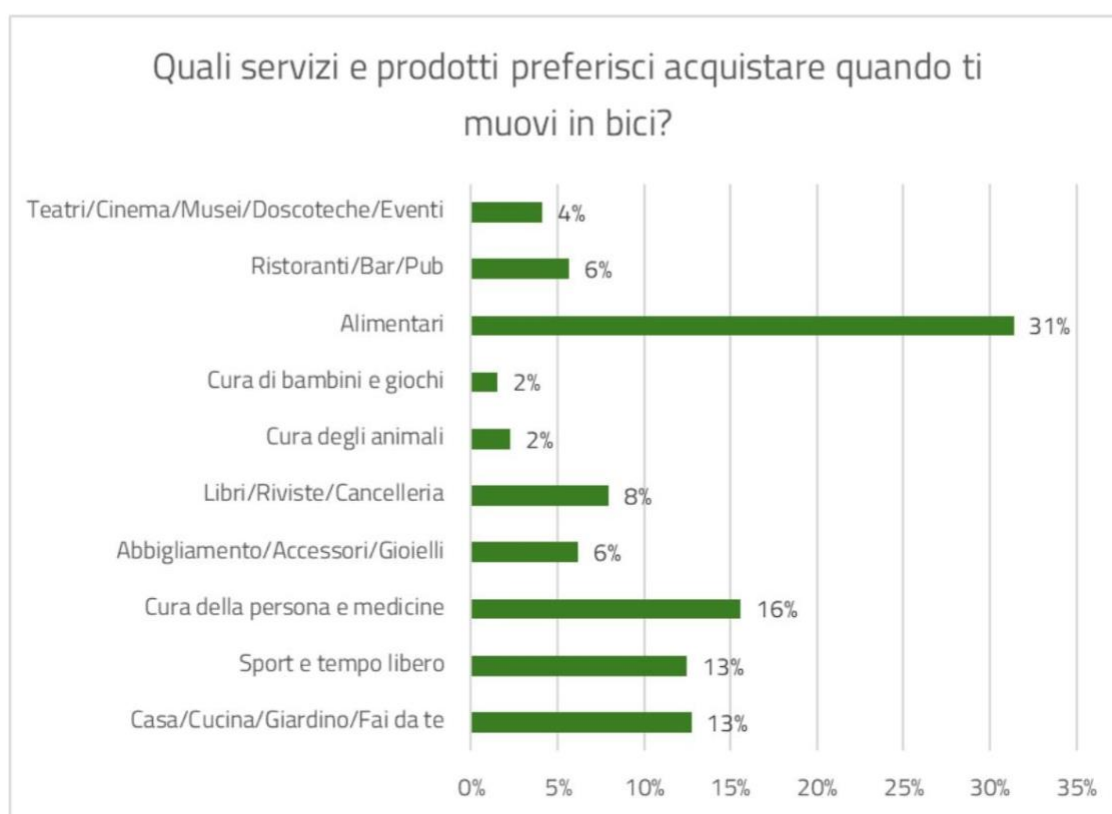


Chart 13: Which services and products do you buy when cycling?
(theatres/cinemas/museums/discos/events, restaurants/bar/pub, grocery, childcare, petcare, books, clothes, healthcare, sport, homeware)

Based on the following analyses and considerations, we can highlight how, even without financial incentives, the sample of users analyzed had an average score of 4.2 on a Likert scale of 1 to 5, where 1 indicates the user is unlikely to continue cycling and 5 indicates the user is extremely likely to continue choosing the bicycle as a mode of transportation. This indicates a significant behavioral shift in the urban areas where Pin Bike projects are active. This finding is undoubtedly linked to the overall rating of the projects and initiatives, approximately 4.7, and to the fact that 98% of users would participate in a Pin Bike initiative again.

These evaluations make it possible to have a better general perception of the bicycle and of alternative systems to the use of the private car as a more sustainable way of getting around the city. More than 50% of users, following the experience with Pin Bike, see this perception changed positively and in general, always considering a Likert scale, the overall index recorded by the sample is

favorable and equal to 3.43, indicating that the majority of users perceive, following incentive initiatives, the bicycle as a "new" mode of urban transport.

Conclusions

The analysis conducted on the "GRINS SIAMUS" survey and the data collected through the questionnaire administered to Pin Bike users in the nine sample cities highlighted a coherent set of results that clearly outline the system's impact on sustainable urban mobility. The adopted methodology, based on a mixed stated preference (SP) and revealed preference (RP) approach, allowed us to understand both the explicit motivations and implicit behavioral dynamics underlying modal choices.

Below are some key points that emerged from the proposed and analyzed questionnaire:

- User profile: the sample, with an average age of 47 and a male prevalence (52%), is representative of the active and working-age urban population, with a significant segment of users who were already cyclists before the project (43%) and a significant share coming from the use of private cars (25%).
- Change in motivations for choice: following the Pin Bike experience, there was an increase in motivations related to environmental sustainability (from 17% to 31%) and economic convenience (from 16% to 27%), compared to a decrease in the importance attributed to speed and convenience.
- Usage behaviors: 75% of users cycle daily or at least four days a week; most travel on average less than 30 minutes, with an average distance of 7.51 km each way.
- Perception of infrastructure: the average rating of simplicity, convenience and safety stands at 4.1 out of 5, with appreciation for the presence of cycle parking and parking services, but also reports of maintenance deficiencies and infrastructural obstacles.
- Shopping by bicycle and local trade: 43% make purchases only out of necessity while on the move, while 26% use their bicycle specifically for shopping, especially for food, indicating a potential positive impact on local trade.

- Lasting behavioral impact: the propensity to continue cycling even without financial incentives reaches an average value of 4.2 out of 5; 98% of users say they are willing to participate in a Pin Bike initiative again, while the perception of the bicycle as an alternative to the car improves significantly (average index 3.43).

The set of data collected highlights that the introduction of a soft incentive system such as Pin Bike has significantly impacted mobility habits, generating a significant increase in bicycle use not only for routine travel but also for everyday activities and shopping. The project encourages the transition to active modes, fosters environmental awareness, and helps reduce reliance on private cars. Although travel speed does not significantly change, a shift in user priorities has been observed, favoring environmental and economic criteria over purely performance-related ones.

From an academic perspective, the analysis provides an original contribution to the literature on behavioral change models in sustainable urban mobility, integrating quantitative and qualitative data collected in heterogeneous urban contexts. For public administrations, the results offer operational guidance on how to structure effective incentive initiatives, maximizing user engagement and ensuring the continued adoption of sustainable modes beyond the

Project duration. For mobility companies, the developed evaluation models represent a useful tool for measuring the socio-environmental returns of interventions and for developing integrated services that combine technology, incentives, and infrastructure.

In summary, the GRINS SIAMUS project demonstrates that an integrated approach of certification, gamification, and economic incentives can be a strategic element in planning sustainable mobility policies, generating measurable environmental, social, and economic benefits and laying the foundations for replicability and scalability in different urban contexts.

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Signature of the Legal Representative

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